Jakob Varenbud – 79236

Interação Pessoa-Máquina 2026/2027

Lab Class 1 Assignment

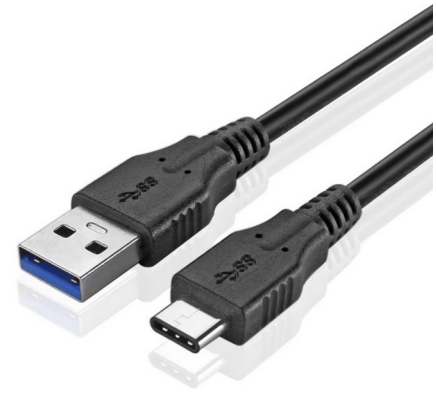
For this assignment I looked at a few interfaces I run into in everyday life – two I think are done well, and two that annoy me almost every time I use them. For each one I describe what it is meant to do, why it works or doesn't, and for the bad ones what I would change.

Good examples of UI

USB-C connectors

USB-C is one of those designs you only appreciate once you've suffered the alternative. The old USB-A plug had a right way up and a wrong way up, but no clear way to tell which was which, so almost everyone flipped it two or three times before it went in. USB-C is symmetrical – it fits both ways round, so there is no wrong orientation at all.

My focus in my master's is Industry 4.0, so I plug in cables constantly, and the difference is real: the task went from “look, guess, flip, try again” to simply pushing it in. The design removes the possibility of the error instead of just warning you about it, which is the strongest kind of error prevention – you can't get it wrong because there is no wrong option left. It solves the problem at the level of the physical shape, not with instructions, and the best fix for a mistake is making it impossible in the first place.



Contactless card terminal

Paying by tapping a card or phone on a terminal is something I do several times a day, and it is a nice small example of good feedback. The moment the payment is accepted the terminal beeps and the screen shows a green tick; if it is declined, you get a different beep and a red cross instead.

The useful part is that the result is signalled in more than one way at the same time – a sound and a colour. In a shop you are often not looking straight at the screen, maybe already putting your card away, and the beep still tells you it worked, so you never have to ask the cashier “did it go through?”. The tap itself is also quick: no inserting, usually no PIN for small amounts, basically one gesture. A common, slightly nervous moment is answered instantly and clearly.



Bad examples of UI

Toaster dial from 1 to 4

Almost every toaster has a dial that goes from 1 to 4 (or some other number), and I still don't really know what those numbers mean. They aren't minutes, and they aren't a browning level you can picture in advance – it's just an abstract scale. So every time I use a toaster I don't know, I end up guessing: pick a number, look at the result, adjust next time.

The control gives you no idea of what it will do and no feedback until the toast is already finished – by then it's too late, and the bread is either barely warm or burnt. The numbers also mean different things on different toasters, so what you learned on one doesn't carry over. To improve it I would label the dial with the outcome instead of numbers (light, medium, dark) and add a progress indicator, like a bar or a changing light, so you can see how far along it is and stop it early if you need to.



Stove controls and their layout

A stovetop usually has four burners arranged in a square, but the control knobs are often lined up in a single row along the front. Nothing about that row tells you which knob belongs to which burner, so you either have to read a tiny diagram next to each knob, or just turn one and watch which ring starts to glow.

This is a textbook case of poor mapping: the layout of the controls doesn't match the layout of the thing they control. Because the connection is arbitrary, I regularly switch on the wrong burner in my flat in Portugal, which isn't just annoying but can be dangerous if something is already sitting on that plate. The fix is well known: arrange the knobs in the same spatial pattern as the burners, even roughly, so the mapping is obvious without any labels at all.

